

I.C. ENGINE

Definition and Basic Components of an Engine, Construction and Working of Two-Stroke and Four-Stroke Spark and Compression Ignition Engines, Comparison Between Two-Stroke and Four-Stroke, SI and CI Engines, Types of fuels.

‡ Definition and Basic Components of an Engine

1. HEAT ENGINES: A heat engine is a machine, which converts heat energy into mechanical energy. The combustion of fuel such as coal, petrol, diesel generates heat. This heat is supplied to a working substance at high temperature. By the expansion of this substance in suitable machines, heat energy is converted into useful work. Heat engines can be further divided into two types:

- (i) External combustion
- (ii) and Internal combustion.

Classification of Heat Engines

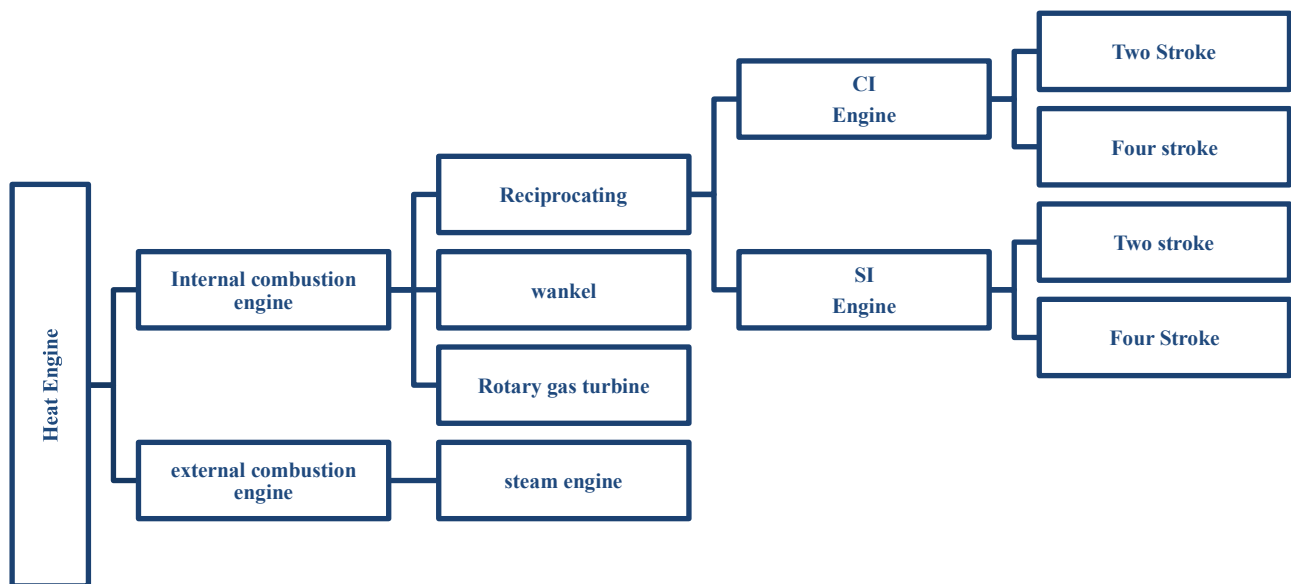


Figure 1: Classification of I.C Engine

1.1 External combustion engine: The working medium, the steam, is generated in a boiler, located outside the engine and allowed into the cylinder to operate the piston to do mechanical work.

1.2 Internal combustion engine : In internal combustion engine, the combustion of fuel takes place inside the engine cylinder and heat is generated within the cylinder. This heat is added to the air inside the cylinder and thus the pressure of the air is increased tremendously. This high-pressure

air moves the piston which rotates the crank shaft and thus mechanical work is done. Most of the internal combustion engines are reciprocating engines with a piston that reciprocate back and forth in the cylinder.

1.3 Components of I.C Engine: Internal combustion engine consists of a number of parts which are given below.

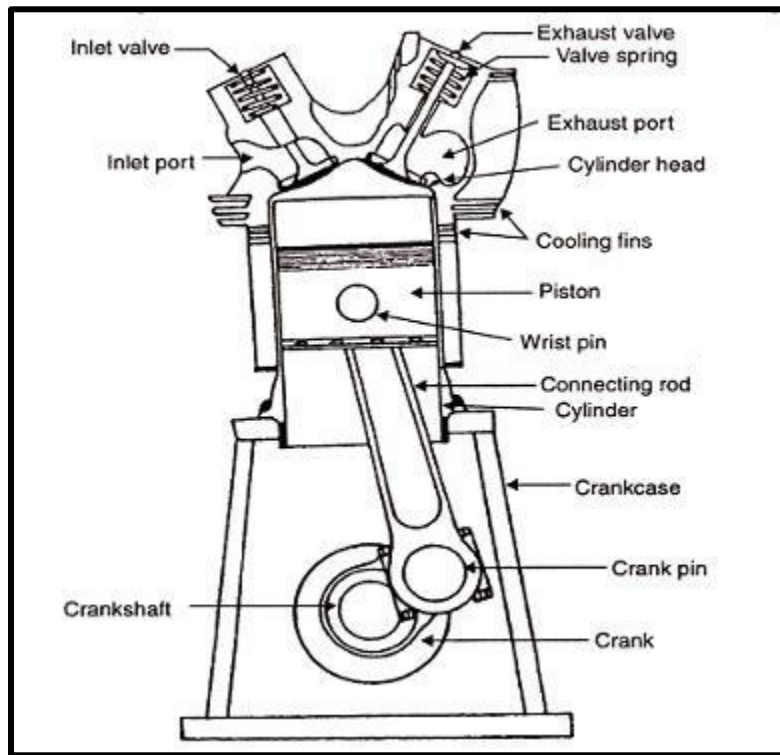


Figure 2. I.C Engine components

- a. Cylinder:** It is a part of the engine which confines the expanding gases and forms the combustion space. It is the basic part of the engine. It provides space in which piston operates to suck the air or air-fuel mixture. The piston compresses the charge, and the gas is allowed to expand in the cylinder, transmitting power for useful work. Cylinders are usually made of high-grade cast iron.
- b. Cylinder block:** It is the solid casting body which includes the cylinder and water jackets (cooling fins in the air-cooled engines).
- c. Cylinder head:** It is a detachable portion of an engine which covers the cylinder and includes the combustion chamber, spark plugs or injector and valves.
- d. Cylinder liner or sleeve:** It is a cylindrical lining either wet or dry type which is inserted in the cylinder block in which the piston slides. Liners are classified as: (1) Dry liner and (2) Wet liner. Dry liner makes metal to metal contact with the cylinder block casing. wet liners come in contact with the cooling water, whereas dry liners do not come in contact with the cooling water.

e. **Piston:** It is a cylindrical part closed at one end which maintains a close sliding fit in the engine cylinder. It is connected to the connecting rod by a piston pin. The force of the expanding gases against the closed end of the piston, forces the piston down in the cylinder. This causes the connecting rod to rotate the crankshaft. Cast iron is chosen due to its high compressive strength. Aluminum and its alloys preferred mainly due to its lightness.

f. **Head (Crown) of piston:** It is the top of the piston. Skirt: It is that portion of the piston below the piston pin which is designed to adsorb the side movements of the piston.

g. **Piston ring:** It is a split expansion ring, placed in the groove of the piston. They are usually made of cast iron or pressed steel alloy. The function of the ring are as follows: a) It forms a gas tight combustion chamber for all positions of piston. b) It reduces contact area between cylinder wall and piston wall preventing friction losses and excessive wear. c) It controls the cylinder lubrication. d) It transmits the heat away from the piston to the cylinder walls. Piston rings are of two types: (1) Compression ring and (2) Oil ring

h. **Compression ring:** Compression rings are usually plain, single piece and are always placed in the grooves of the piston nearest to the piston head. They prevent leakage of gases from the cylinder and helps increasing compression pressure inside the cylinder.

i. **Oil ring:** Oil rings are grooved or slotted and are located either in lowest groove above the piston pin or in a groove above the piston skirt. They control the distribution of lubrication oil in the cylinder and the piston.

j. **Piston Pin:** It is also called wrist pin or gudgeon pin. Piston pin is used to join the connecting rod to the piston.

k. **Connecting rod:** It is special type of rod, one end of which is attached to the piston and the other end to the crankshaft. It transmits the power of combustion to the crankshaft and makes it rotate continuously. It is usually made of drop forged steel.

l. **Crankshaft:** It is the main shaft of an engine which converts the reciprocating motion of the piston into rotary motion of the flywheel. Usually, the crankshaft is made of drop forged steel or cast steel. The space that supports the crankshaft in the cylinder block is called main journal, whereas the part to which connecting rod is attached is known as crank journal. Crankshaft is provided with counterweights throughout its length to have counterbalance of the unit.

m. **Camshaft:** It is a shaft which raises and lowers the inlet and exhaust valves at proper times. Camshaft is driven by crankshaft by means of gears, chains or sprockets. The speed of the camshaft is exactly half the speed of the crankshaft in four stroke engine. Camshaft operates the ignition timing

mechanism, lubricating oil pump and fuel pump. It is mounted in the crankcase, parallel to the crankshaft.

2. IC ENGINE TERMINOLOGIES

Bore- Bore is the diameter of the engine cylinder.

Top dead centre (TDC) and Bottom Dead centre (BDC) - TDC is the position where the piston is at its highest point in the cylinder, closest to the cylinder head, while BDC is the position where the piston is at its lowest point in the cylinder, farthest from the cylinder head.

Stroke - It is the linear distance travelled by the piston from Top dead center (TDC) to Bottom dead center (BDC).

Stroke-bore ratio -The ratio of length of stroke (L) and diameter of bore (D) of the cylinder is called stroke-bore ratio (L/D). In general, this ratio varies between 1 to 1.45 and for tractor engines, this ratio is about 1.25.

Swept volume - It is the volume ($A \times L$) displaced by one stroke of the piston where A is the cross sectional area of piston and L is the length of stroke
Top dead center - When the piston is at the top of its stroke, it is said to be at the top dead center (TDC), Bottom dead center - when the piston is at the bottom of its stroke, it is said to be at its bottom dead center (BDC).

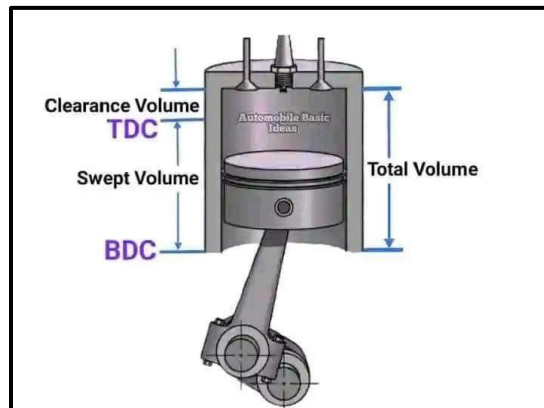


Figure 3. IC Engine Terminologies

Compression ratio: It is the ratio of the volume of the cylinder at the beginning of the compression stroke to that at the end of compression stroke, i.e. ratio of total cylinder volume to clearance volume. The Compression ratio of diesel engine varies from 16:1 to 22:1 and that of carburetor type engine (spark ignition engine) varies from 8:1 to 12:1.

3. FOUR STROKE CYCLE ENGINE (DIESEL/ PETROL ENGINE)

In four stroke cycle engines the four events namely suction, compression, power and exhaust take place inside the engine cylinder. The four events are completed in four strokes of the piston (two

revolutions of the crank shaft). This engine has got valves for controlling the inlet of charge and outlet of exhaust gases. The opening and closing of the valve is controlled by cams, fitted on camshaft. The camshaft is driven by crankshaft with the help of suitable gears or chains. The camshaft runs at half the speed of the crankshaft. The events taking place in I.C. engine are as follows:

1. Suction stroke
2. Compression stroke
3. Power stroke
4. Exhaust stroke

3.1 Four stroke petrol Engines or SI Engine

The four-stroke petrol engine operates based on four systems: suction, compression, power, and exhaust. The details of all four systems are given below. The figure 3 shows the working step for four stroke SI engine

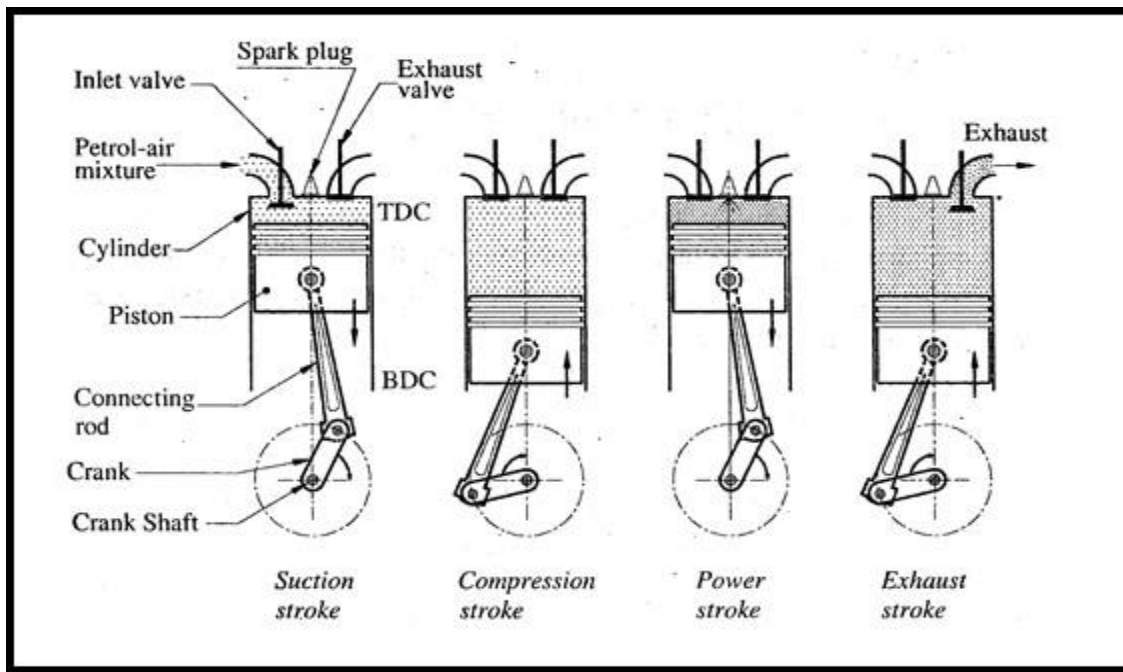


Figure 4. Steps of four stroke S.I engine

i. Suction stroke

During suction stroke inlet valve opens and the piston moves downward. A mixture of air and fuel are drawn inside the cylinder. The exhaust valve remains in closed position during this stroke. The pressure in the engine cylinder is less than atmospheric pressure during this stroke.

ii. Compression stroke

During this stroke the piston moves upward. Both valves are in closed position. The charge taken in the cylinder is compressed by the upward movement of piston. A mixture of air and fuel is compressed in the cylinder, as in case of petrol engine, the mixture is ignited by a spark plug.

iii. Power stroke

After ignition of fuel, tremendous amount of heat is generated, causing very high pressure in the cylinder which pushes the piston downward. The downward movement of the piston at this instant is called power stroke. The connecting rod transmits the power from piston to the crank shaft and crank shaft rotates. Mechanical work can be taped at the rotating crank shaft. Both valves remain closed during power stroke.

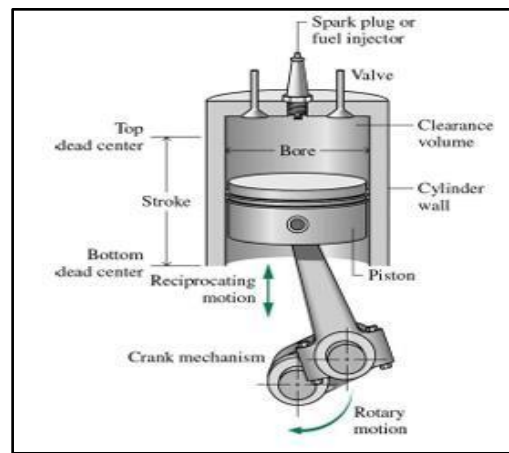


Figure 5. Details of power stroke

iv. Exhaust stroke

During this stroke piston moves upward. Exhaust valve opens and exhaust gases go out through exhaust valves opening. All the burnt gases go out of the engine and the cylinder becomes ready to receive the fresh charge. During this stroke inlet valve remains closed. Thus, it is found that out of four strokes, there is only one power stroke and three idle strokes in four stroke cycle engines.

The power stroke supplies necessary momentum for useful work.

3.2 Two stroke petrol Engines or SI Engine

In two stroke cycle engines, the whole sequence of events i.e., suction, compression, power and exhaust are completed in two strokes of the piston i.e. one revolution of the crankshaft. There is no valve in this type of engine. Gas movement takes place through holes called ports in the cylinder. The crankcase of the engine is airtight in which the crankshaft rotates.

a. Upward stroke of the piston (Suction + Compression)

When the piston moves upward it covers two of the ports, the exhaust port and transfer port, which are normally almost opposite to each other. This traps the charge of air- fuel mixture drawn already in to the cylinder. Further upward movement of the piston compresses the charge and also uncovers the suction port. Now fresh mixture is drawn through this port into the crankcase. Just before the end of this stroke, the mixture in the cylinder is ignited by a spark plug. Thus, during this stroke both suction and compression events are completed.

b. Downward stroke (Power + Exhaust)

Burning of the fuel rises the temperature and pressure of the gases which forces the piston to move down the cylinder. When the piston moves down, it closes the suction port, trapping the fresh charge drawn into the crankcase during the previous upward stroke. Further downward movement of the piston uncovers first the exhaust port and then the transfer port. Now fresh charge in the crankcase moves into the cylinder through the transfer port driving out the burnt gases through the exhaust port. Special shaped piston crown deflects the incoming mixture up around the cylinder so that it can help in driving out the exhaust gases, known as scavenging. During the downward stroke of the piston power and exhaust events are completed.

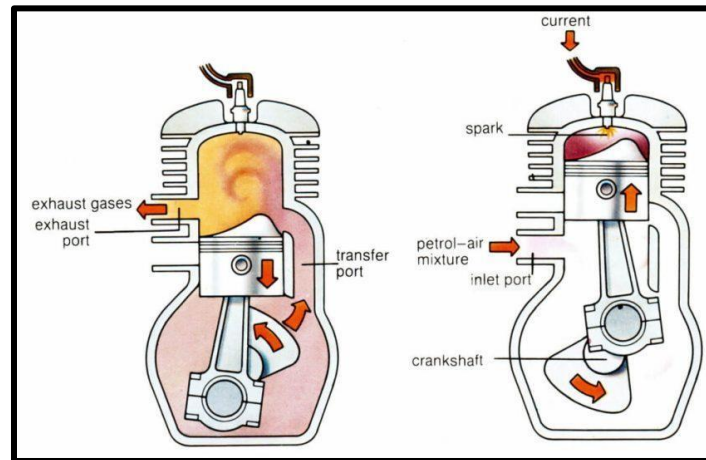


Figure 6. Two stroke petrol engine construction

4. Four stroke Diesel Engines or CI Engine

The following steps to be followed for the working of four stroke diesel engine.

a. Suction Stroke (First Stroke of the piston) •

Piston moves from TDC to BDC

- Inlet valve is opened, and the exhaust valve is closed.
- The pressure inside the cylinder is reduced below the atmospheric pressure.

- Fresh air from the atmosphere is sucked into the engine cylinder through air cleaner and inlet valve.

b. Compression Stroke (second stroke of the piston)

- Piston moves from BDC to TDC.
- Both inlet and exhaust valves remain closed.
- The air drawn during suction stroke is compressed to a high pressure and temperature.

c. Working or power or expansion stroke (Third stroke of the piston)

- The burning gases (products of combustion) expand rapidly.
- The burning gases pushes the piston to move downward from TDC to BDC.
- This movement of piston is converted into rotary motion of the crank shaft through connecting rod.
- Both inlet and exhaust valves remain closed

d. Exhaust Stroke (Fourth stroke of the piston)

- Piston moves from BDC to TDC.
- Exhaust valve is opened and the inlet valve remains closed.
- The burnt gases are forced out to the atmosphere through the exhaust valve. (some of the burnt gases stay in the clearance volume of the cylinder)
- The exhaust valve closes shortly after reaches TDC.
- The inlet valve opens slightly before TDC and the cylinder is ready to receive fresh air to start a new cycle.

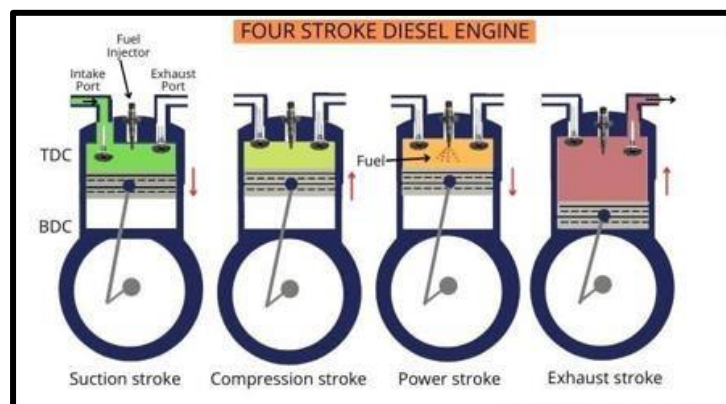


Figure 7: working of four stroke diesel engines

4.1 Two stroke Diesel Engines or CI Engine A.

First Stroke (Upward Stroke of the piston)

(a) Compression and inductance:

- The piston moves upwards from Bottom Dead Centre (BDC) to Top Dead Centre (TDC).
- Both transfer and exhaust ports are covered.
- Air which is transferred already into the engine cylinder is compressed by moving piston.
- The pressure and temperature of the air increases.
- At the same time, fresh air is admitted into the crankcase through the plate valve (reed valve)

(b) Ignition and inductance.

- Piston almost reaches the top dead centre.
- The fuel is injected into the hot compressed air inside the cylinder. The fuel mixes with hot air and burns.
- The admission of fresh air into the crankcase continues till the piston reaches the top centre

B. Second Stroke (Downward Stroke of the piston)

(c) Expansion and crank case compression:

- •The burning gases expand in the cylinder.
- •Burning gases force the piston to move down. Thus useful work is obtained.
- •At the same time, the air in the crank case is compressed by the movement of the piston.
- All the ports and the plate valve are in closed position.

(d) Exhaust and Transfer

- At the end of expansion, the exhaust port is uncovered.
- The burnt gases escape to the atmosphere through the exhaust port.
- Transfer port is also uncovered shortly after the exhaust port is opened.
- The partially compressed air from crank case enters the cylinder through transfer port. products from the cylinder
- This air is deflected upwards by the deflected shape of the piston.
- Thus, the entering air helps in forcing out the combustion products, the scavenging process
- The plate valve remains during this period

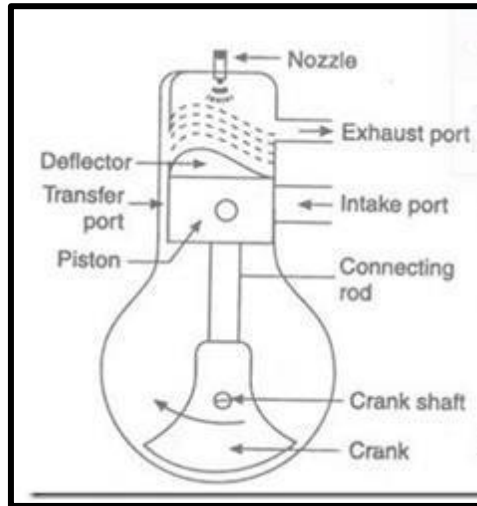


Figure 8: Working of two stroke diesel engines.

5. Comparison Between SI and CI Engines

Aspect	Diesel Engine	Petrol Engine
1. Components	No carburetor, ignition coil, or spark plug.	Has carburetor, ignition coil, and spark plug.
2. Compression Ratio	14:1 to 22:1	8:1 to 12:1
3. Fuel Used	Diesel oil	Petrol (gasoline) or power kerosene
4. Suction Stroke	Only air is sucked into the cylinder.	Air–fuel mixture is sucked into the cylinder.
5. Fuel System	Has fuel injection pump and injector.	No injector; has carburetor and ignition coil.
6. Combustion Process	Fuel is injected into hot compressed air; self-ignition occurs due to heat of compression.	Air–fuel mixture is compressed and ignited by an electric spark.
7. Thermal Efficiency	32% – 38% (higher).	25% – 32% (lower).
8. Engine Weight (per HP)	High (heavier).	Comparatively low (lighter).
9. Operating Cost	Low (diesel is cheaper, efficiency higher).	High (fuel cost and lower efficiency).
10. Compression Pressure & Temperature	35–45 kg/cm ² ; ~500 °C.	6–10 kg/cm ² ; ~260 °C.

6. Comparison Between two stroke and four stroke Engine

Aspect	Two-Stroke Engine	Four-Stroke Engine
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1. Power Stroke	One power stroke in every revolution of crankshaft.	One power stroke in every two revolutions of crankshaft.
2. Number of Strokes per Cycle	2 strokes (one up and one down).	4 strokes (suction, compression, power, exhaust).
3. Power Output	Higher (because one power stroke per revolution).	Lower (because one power stroke per two revolutions).
4. Size/Weight	Lighter and more compact for same power.	Heavier and bulkier for same power.
5. Efficiency	Less fuel efficient (due to incomplete combustion and loss of fuel with exhaust).	More fuel efficient (complete combustion and better fuel utilization).
6. Durability/Life	Shorter engine life (more wear and tear due to high running speed & lubrication issues).	Longer engine life (less wear, better lubrication).
7. Maintenance	Requires more frequent maintenance.	Requires less frequent maintenance.
8. Lubrication	Uses lubricating oil mixed with fuel (totalloss lubrication).	Uses separate lubrication system (sump lubrication).
9. Cost	Cheaper in cost.	More expensive.
10. Applications	Motorcycles, scooters, mopeds, auto rickshaws, outboard motors, small engines.	Cars, buses, trucks, tractors, heavy vehicles, power plants.

7. Types of fuels

Internal combustion (IC) engines primarily utilize hydrocarbon-based fuels like gasoline, diesel, and natural gas. These fuels, when burned, release chemical energy that is converted into mechanical energy to power the engine. Other fuels, including biofuels like ethanol and biodiesel, as well as hydrogen, can also be used in IC engines.

7.1 Types of Fuels for IC Engines

A. Liquid Fuels

(Most commonly used in IC engines)

1. Petrol (Gasoline):

Used in spark-ignition (SI) engines (cars, bikes, scooters).
Volatile, easy to vaporize, ignited by spark plug.

2. Diesel:

Used in compression-ignition (CI) engines (trucks, buses, tractors, generators).
Less volatile, ignites due to heat of compression.

3. Kerosene:

Sometimes used in stationary/aviation engines. **4. Alcohol Fuels (Ethanol, Methanol):**
Can be blended with petrol (e.g., E10, E20).
Renewable & cleaner.

5. Biodiesel:

Made from vegetable oils/animal fats, alternative to diesel.

B. Gaseous Fuels

6. CNG (Compressed Natural Gas):

Mostly methane; used in SI engines with modifications.

7. LPG (Liquefied Petroleum Gas):

Propane + Butane; used in taxis, autorickshaws, buses.

8. Biogas:

Methane from organic waste; used in rural IC engines.

9. Hydrogen:

Clean fuel, produces only water vapor, used in experimental engines.

10. Producer Gas, Coal Gas, Water Gas:

Industrial/older applications.

C. Alternative / Future Fuels

- **Hydrogen fuel (H_2):** For fuel cell vehicles or direct IC combustion.
- **Synthetic Fuels:** Derived from coal/gas.
- **Biofuels:** Ethanol, biodiesel blends.
- **Algae-based Fuels:** Under research.

8. Basic properties of fuel in I.C Engine: Ideal internal combustion (IC) engine fuels have high energy density, good combustion qualities like proper vaporization and a suitable knock rating (octane for gasoline, cetane for diesel), thermal stability, and low tendency to form deposits or corrode engine components. They must also be easy and safe to handle, store, and transfer, with low toxicity and minimal environmental impact from their combustion byproducts.

† Properties of a Good Fuel

a. Calorific Value (CV)

- Amount of heat energy released when 1 kg (solid/liquid) or 1 m³ (gas) of fuel is completely burned.
- Measured in kJ/kg or kJ/m³.
- A good fuel should have high calorific value (more energy per unit).

b. Ignition Temperature

- Minimum temperature at which a fuel starts burning.
- Should be moderate:
- Not too low (to avoid fire hazards).
- Not too high (so that ignition is easy).

c. Volatility

- Ability of a fuel to vaporize.
- Petrol must be volatile (easy mixing with air in SI engines).
- Diesel requires controlled volatility (to avoid knocking).

d. Flash Point

- Lowest temperature at which fuel gives off enough vapor to ignite momentarily.
- Petrol: ~40 °C (low), Diesel: ~60–80 °C (safer).

e. Fire Point

- Slightly higher than flash point, where fuel vapors burn continuously.

f. Viscosity (for liquid fuels)

- Resistance to flow.
- Should not be too high (difficult injection) or too low (causes leakage).

g. Knocking Tendency

- In SI engines → related to Octane Number (higher = better).
- In CI engines → related to Cetane Number (higher = smoother ignition).

h. Sulphur Content

- Should be low, because Sulphur causes corrosion and pollution (SO₂ gas).

i. Ash Content (for solid fuels)

- Residue left after combustion.
- Should be minimal to avoid deposits and maintenance.

j. Storage & Handling

- Fuel should be easy and safe to store, transport, and handle.

k. Cost & Availability

- Economical and readily available near point of use.

Electric vehicles and hybrid electric vehicles

Working principle and components of an EV, advantages and disadvantages of EVs. working of hybrid electric vehicles, series and parallel types of HEV, degree of hybridization, advantages and disadvantages of HEV.

Electric vehicle:

Electric vehicles (EVs) use electricity as their primary fuel or to improve the efficiency of conventional vehicle designs. EVs include all-electric vehicles, also referred to as battery electric vehicles (BEVs), and plug-in hybrid electric vehicles (PHEVs). In colloquial references, these vehicles are called electric cars, or simply EVs, even though some of these vehicles still use liquid fuels in conjunction with electricity. EVs are known for providing instant torque and a quiet driver experience.

Evolution of Electric Vehicle (EV) Development

1. Early Developments (1769–1890)

The origins of vehicle technology trace back to the invention of the steam engine in 1769, which marked the beginning of mechanized transport. In 1806, the first internal combustion engine (ICE) powered by fuel gas was developed. Later, in 1839, the first successful gas-driven engine was created, paving the way for gasoline-powered cars. By 1893, the first gasoline-powered car was introduced, offering improved range and fuelling convenience.

2. The Golden Age of Electric Vehicles (1890–1924)

This period is regarded as the Golden Age of EVs. EVs became popular due to their quiet operation, ease of driving, and absence of exhaust emissions. By 1912, EVs reached peak production and were seen as strong competitors to gasoline-powered vehicles. However, limitations of battery technology, high cost, and restricted driving range gradually reduced their attractiveness.

3. Domination of Gasoline Cars (1924–1960)

Between 1924 and 1960, gasoline-powered cars became dominant, leading to the near disappearance of EVs. Mass production techniques, greater range, and fueling convenience made gasoline cars more accessible. EVs lost market share completely during this time.

4. Revival and Early Prototypes (1960–1980)

From the 1960s onward, environmental concerns and oil crises reignited interest in EVs. Serious prototypes emerged in the 1960s, limited production in the 1970s, and incremental technological advances in the 1980s improved EV reliability.

5. First Generation of Modern EVs (1990–2000)

The 1990s marked the introduction of the first generation of modern EVs, supported by government policies and environmental concerns. However, limited range, long charging times, and high costs restricted widespread adoption.

6. Hybrids and Renewed Interest (2000–Present)

Since the early 2000s, hybrid vehicles played a crucial role in bridging toward full EV adoption. Rising oil prices, technological advancements in batteries, and government incentives brought EVs back into focus.

7. Future Outlook

With rapid advancements in lithium-ion and solid-state batteries, EVs are positioned as the future of sustainable transportation. Global automakers are investing heavily, supported by policies and sustainability goals. Figure 9-10 shows the top electrical vehicle maker in the world as well as India

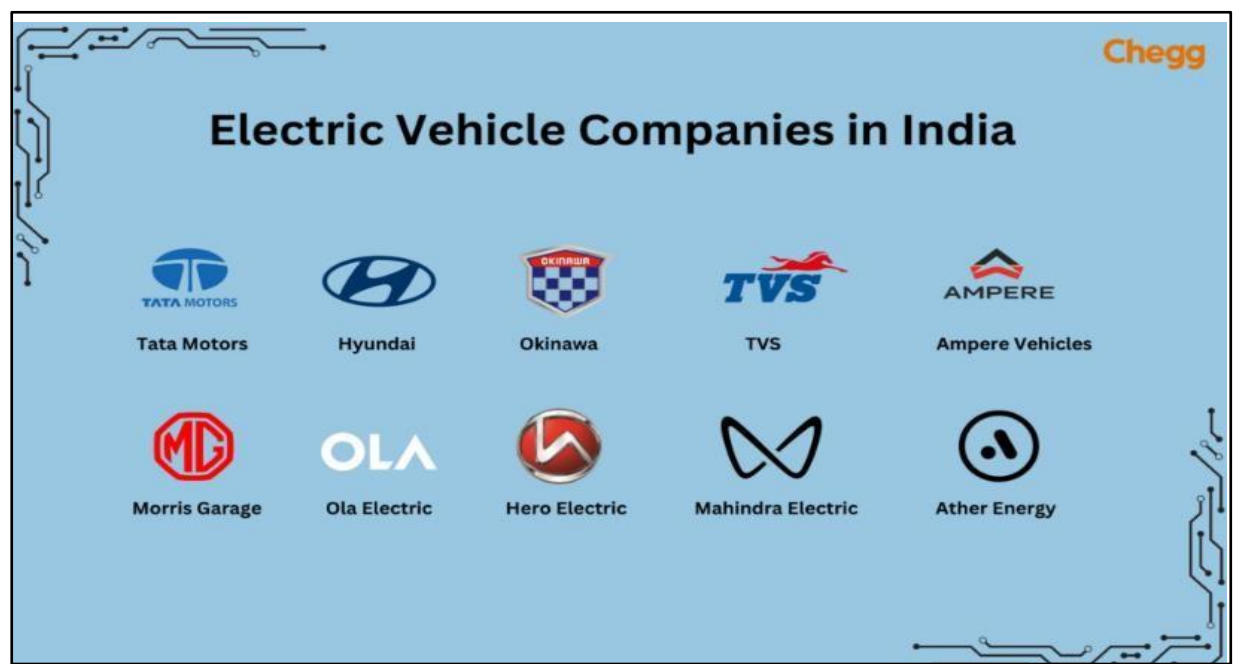


Figure 9 : Indian top electrical vehicle maker

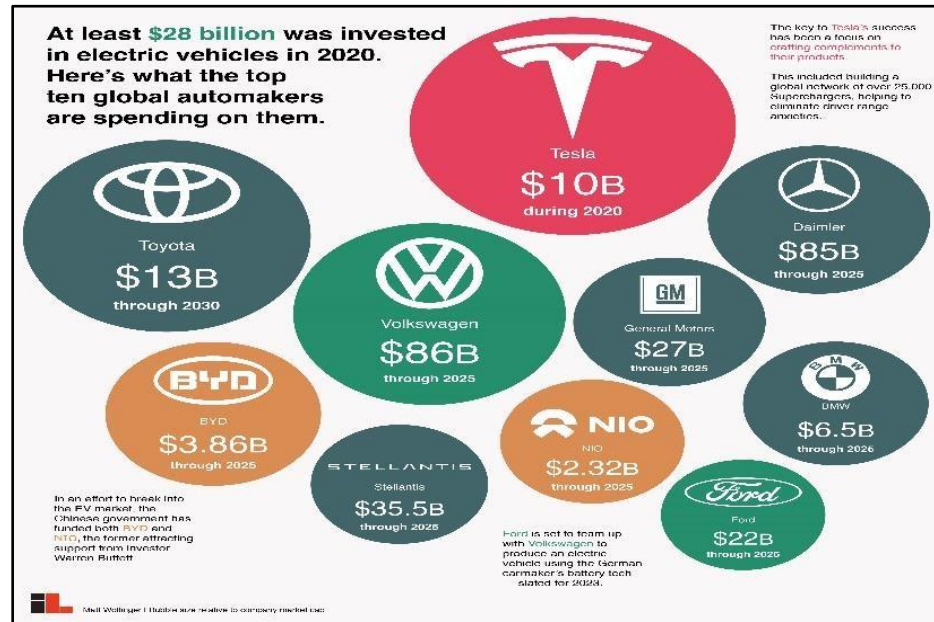


Figure 10 : Global top electrical vehicle maker

Working principle and components of an EV

In the following way the electric car is getting the motion and fulfil the required operation.

- Power is converted from the DC battery to AC for the electric motor
- The accelerator pedal sends a signal to the controller which adjusts the vehicle's speed by changing the frequency of the AC power from the inverter to the motor
- The motor connects and turns the wheels through a cog
- When the brakes are pressed or the electric car is decelerating, the motor becomes an alternator and produces power, which is sent back to the battery. Rotation of the motor rotor rotates the transmission, so the wheels turn and then the car moves. Figure 11 and figure 12 shows the working principle and workflow of the electric vehicle. shows the flow chart of working of electrical vehicle.

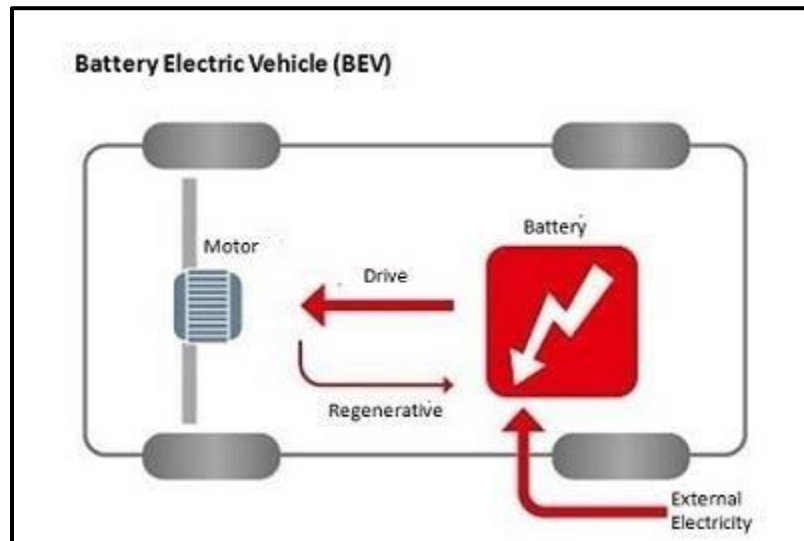


Figure 11. working of electrical vehicle

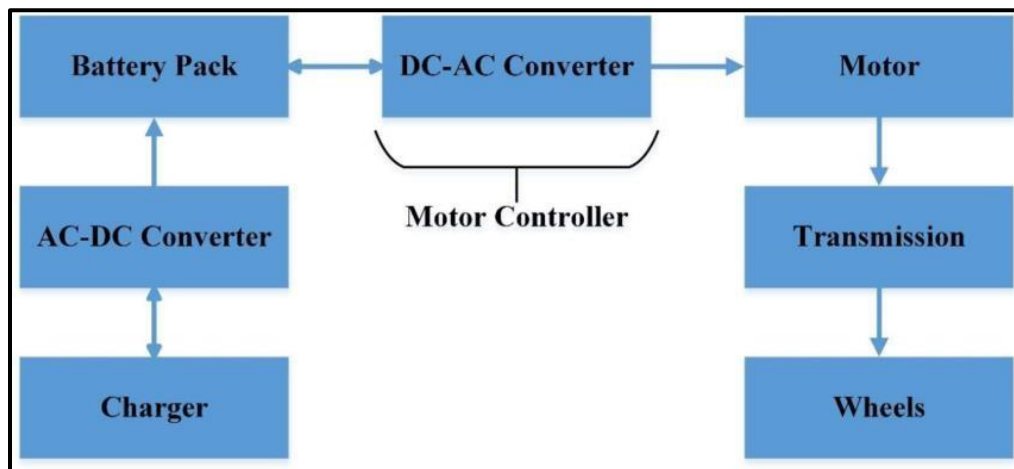


Figure 12. Flow chart of working of electrical vehicle

Examples of BEV

Volkswagen e-Golf, Tesla Model 3, BMW i3, Chevy Bolt, Chevy Spark, Nissan LEAF, Ford Focus Electric, Hyundai Ioniq, Karma Revera, Kia Soul, Mitsubishi i-MiEV, Tesla X, Toyota Rav4.

Components of electrical vehicle

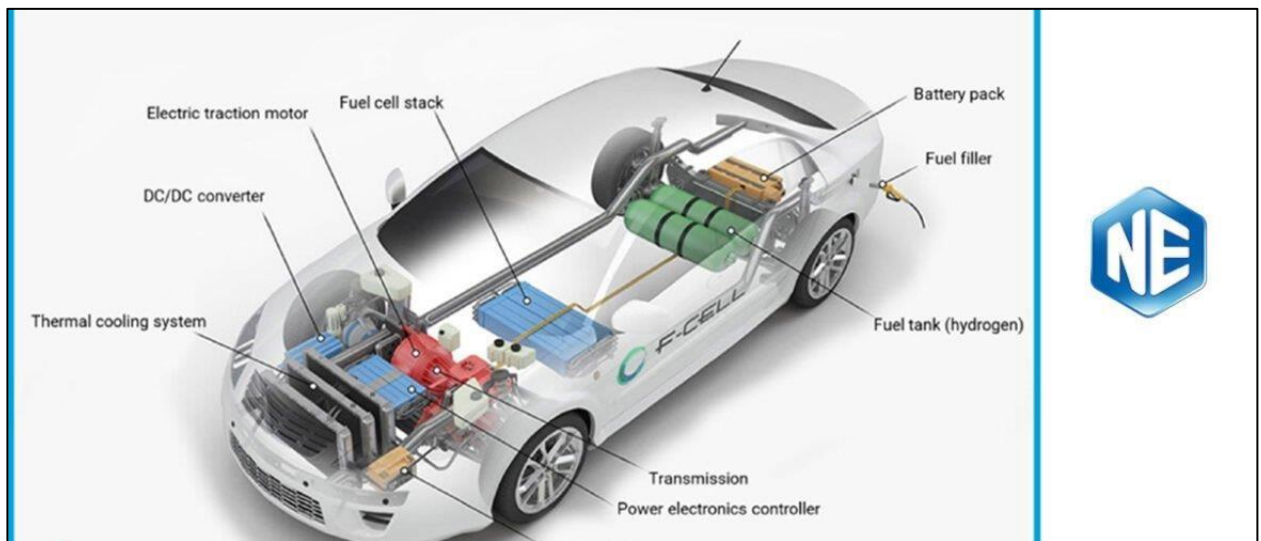


Figure 13. Flow chart of working of electrical vehicle

Battery:

- The battery is the primary power source in an electric vehicle.
- It consists of a large number of electrical cells that store electrical energy.
- The battery is charged by connecting the vehicle to an external power source, such as a home or public charging point.
- The battery's capacity (in kilowatt-hours, kWh) determines the vehicle's range on a single charge.

Electric Motor:

- Instead of the internal combustion engine found in conventional cars, electric vehicles use one or more electric motors.
- When the motor is supplied with electrical energy from the battery, it converts this energy into rotational kinetic energy.
- The electric motor has instant torque, providing quick and smooth acceleration.
- Most electric vehicles use alternating current (AC) motors, but some use direct current (DC) motors for auxiliary components.

Electronic Control Unit (ECU) or Power Control Unit:

- This unit acts as the “brain” of the electric propulsion system.

- It manages the flow of energy between the battery and the motor.
- It controls the motor's speed and torque based on driver input (such as pressing the accelerator pedal).
- It also oversees other functions such as regenerative braking.

Regenerative Braking System:

- This is an important feature in electric vehicles that helps recover some energy during deceleration or braking.
- When the driver takes their foot off the accelerator or applies the brakes, the electric motor turns into a generator.
- The generator converts the vehicle's kinetic energy back into electrical energy and feeds it back into the battery, increasing energy efficiency and range.

Transmission (usually simple):

- In most electric vehicles, the transmission is very simple and often consists of a single fixed reduction gear.
- Due to the electric motor's instant torque and wide speed range, electric vehicles don't require a complex gearbox like internal combustion vehicles.
- This simplifies the drivetrain and reduces maintenance.

Charging System:

- Allows the vehicle to recharge its battery pack from an external power source, such as a wall socket or charging station.

Battery Management System (BMS):

Monitors and manages the battery pack's state of charge, health, and safety.

An electric vehicle draws its power from a rechargeable battery, which powers an electric motor. The electric motor converts this energy into motion, which drives the wheels. The electronic control unit helps manage power flow, while the regenerative braking system recovers some energy during deceleration and braking. This system is much simpler than

the propulsion system in conventional cars, resulting in fewer moving parts and reduced maintenance requirements.

Advantage and disadvantage of electric vehicle

Advantages

1. Environmental Benefits:

EVs produce no direct tailpipe emissions, significantly reducing air pollution and contributing to a smaller carbon footprint.

2. Lower Operating Costs:

Electricity is generally cheaper than gasoline, and EVs require less maintenance because they have fewer mechanical parts, meaning no oil changes or exhaust systems.

3. Quiet and Smooth Ride:

Electric motors provide a quiet and smooth driving experience with instant torque for quick acceleration.

4. Convenient Charging:

Drivers can charge their vehicles at home, which is like starting each day with a full "tank".

5. Government Incentives:

Many governments offer tax credits and other incentives to encourage EV adoption.

6. Technological Advancements:

EVs often incorporate cutting-edge technology, offering advanced automation and a superior driving experience.

Disadvantage

1. High Initial Cost:

EVs typically have a higher purchase price compared to conventional gasoline cars, although this can be offset by lower operating costs and incentives.

2. Limited Driving Range:

While improving, the driving range on a single charge can be a limitation for some drivers, leading to "range anxiety".

3. Charging Infrastructure:

The availability of public charging stations is growing but can still be a challenge, especially in certain areas or during long trips.

4. Longer Charging Times:

Recharging an EV battery takes longer than filling a gas tank, with times varying from minutes to several hours depending on the charger type and battery size.

5. Battery Life and Cost:

While batteries are improving, they have a finite lifespan and can be very expensive to replace.

6. Impact on Power Grid:

Increased EV adoption could strain the electricity grid, potentially negating some of their environmental benefits if the electricity isn't generated from renewable sources.

Working of hybrid electric vehicles, series and parallel types of HEV

Types of Electric Cars

There are 4 (four) types of electric cars, with the following outline:

1. Battery Electric Vehicle (BEV)

2. Hybrid

i. Hybrid Electric Vehicle (HEV) ii. Plug-in Hybrid Electric Vehicle (PHEV)

3. Fuel Cell Electric Vehicle (FCEV)

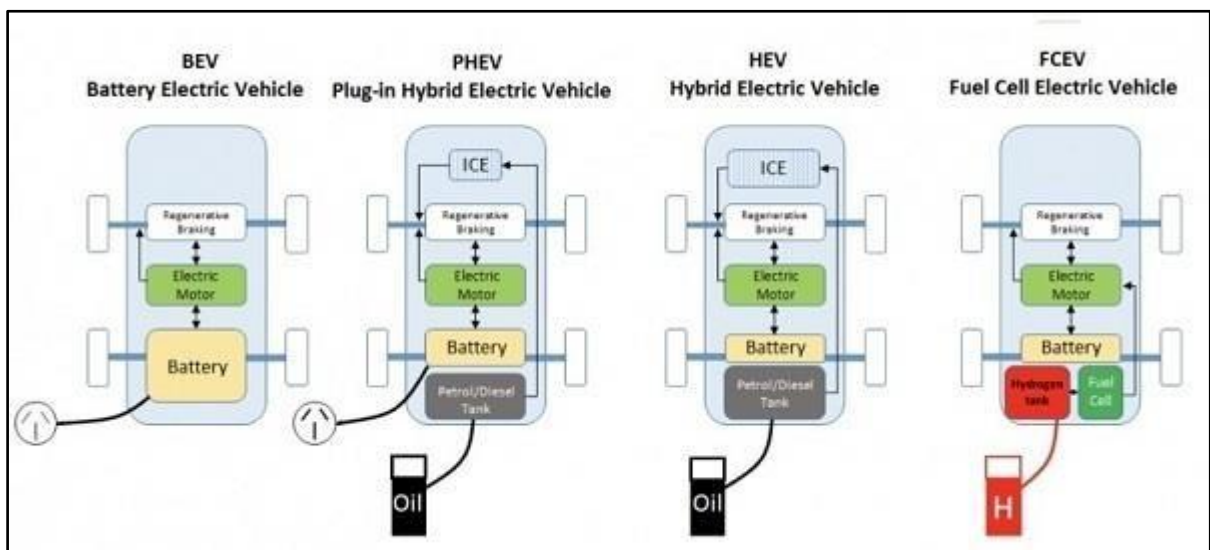


Figure 14. Types of electrical vehicle

Hybrid Electric Vehicle (HEV)

This type of hybrid cars is often called as standard hybrid or parallel hybrid. HEV has both an ICE and an electric motor. In this types of electric cars, internal combustion engine gets energy from fuel (gasoline and others type of fuels), while the motor gets electricity from batteries. The gasoline engine and electric motor simultaneously rotate the transmission, which drives the wheels. The difference between HEV compared to BEV and PHEV is where the batteries in HEV can only charge by the ICE, the motion of the wheels or a combination of both. There is no charging port, so that the battery cannot be recharged from outside of the system, for example from the electricity grid.

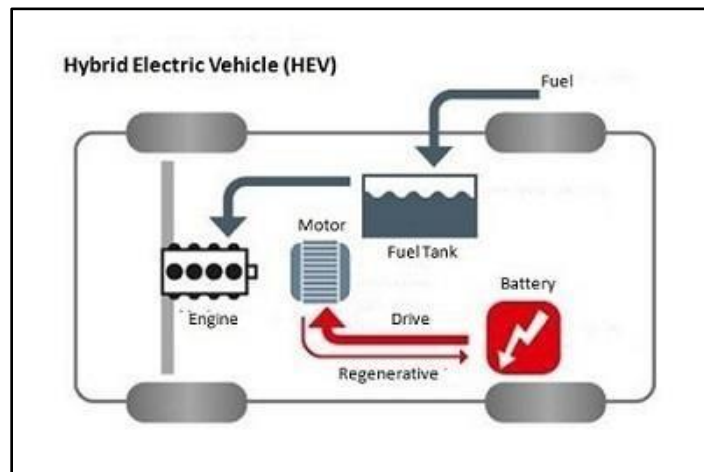


Figure 15. Types of electrical vehicle

Components of HEV

- Engine
- Electric motor
- Battery pack with controller & inverter
- Fuel tank
- Control module

Working Principles of HEV

- Has a fuel tank that supplies gas to the engine like a regular car
- It also has a set of batteries that run an electric motor
- Both the engine and electric motor can turn the transmission at the same time

Examples of HEV

Honda Civic Hybrid, Toyota Prius Hybrid, Honda Civic Hybrid, Toyota Camry Hybrid.

Plug-in Hybrid Electric Vehicle (PHEV)

PHEV is a type of hybrid vehicle that both an ICE and a motor, often called as **series hybrid**. This type of electric cars offers a choice of fuels. This type of electric cars is powered by a conventional fuel (such as gasoline) or an alternative fuel (such as biodiesel) and by a rechargeable battery pack. The battery can be charged up with electricity by plugging into an electrical outlet or electric vehicle charging station (EVCS). **PHEV typically can run in at least two modes:**

- All-electric Mode, in which the motor and battery provide all the car's energy
- Hybrid Mode, in which both electricity and gasoline are employed.

Some PHEVs can travel more than 70 miles on electricity alone.

Architecture and Main Components of PHEV

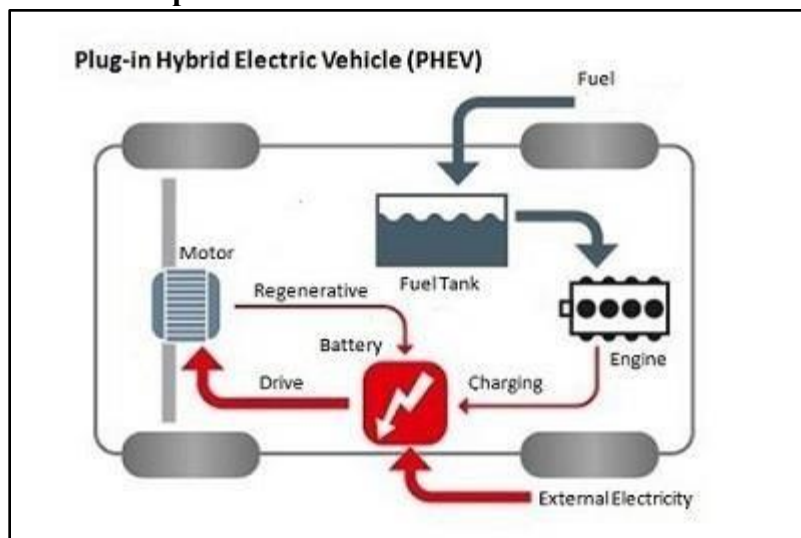


Figure 16. Types of electrical vehicle (PHEV)

Working Principles of PHEV

PHEVs typically start up in all-electric mode and operate on electricity until their battery pack is depleted. Some models shift to hybrid mode when they reach highway cruising speed, generally above 60 or 70 miles per hour. Once the battery is empty, the engine takes over and the vehicle operates as a conventional, non-plug-in hybrid.

In addition to plugging into an outside electric power source, PHEV batteries can be charged by an internal combustion engine or regenerative braking. During braking, the electric motor acts as a generator, using the energy to charge the battery. The electric motor supplements the engine's power; as a result, smaller engines can be used, increasing the car's fuel efficiency without compromising performance.

Examples of PHEV

Porsche Cayenne S E-Hybrid, Chevy Volt, Chrysler Pacifica, Ford C-Max Energi, Ford Fusion Energi, Mercedes C350e, Mercedes S550e, Mercedes GLE550e.

Fuel Cell Electric Vehicle (FCEV)

Fuel Cell Electric Vehicles (FCEVs), also known as fuel cell vehicles (FCVs) or Zero Emission Vehicle, are types of electric cars that employ 'fuel cell technology' to generate the electricity required to run the vehicle. In this type of vehicles, the chemical energy of the fuel is converted directly into electric energy.

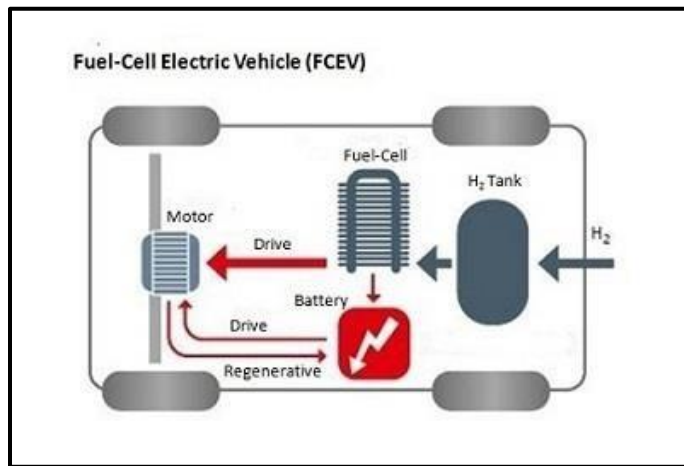


Figure 17 Types of electrical vehicle (FCEV)

Working Principles of FCEV

The working principle of a 'fuel cell' electric car is different compared to that of a 'plugin' electric car. This type of electric cars is because the FCEV generates the electricity required to run this vehicle on the vehicle itself.

Examples of FCEV

Toyota Mirai, Hyundai Tucson FCEV, River simple Rasa, Honda Clarity Fuel Cell, Hyundai Nexo.

Series and parallel types of HEV Series

types of HEV:

Series Hybrid Vehicle – Working Principle

1. Fuel Supply & Engine Operation

- The fuel tank supplies fuel to the internal combustion engine (ICE).
- The engine does not drive the wheels directly; instead, it operates primarily to run at its most efficient speed and load.

2. Generator & Battery Charging

- The engine drives a generator, which converts mechanical energy $\rightarrow$ electrical energy.
- The generated electricity is used to charge the battery pack via a charger.

3. Battery & Power Electronics

- The battery pack stores energy and supplies it to the power controller.
- The controller regulates voltage and current supplied to the electric traction motor.
- Depending on the motor type, an inverter circuit is used to convert DC $\rightarrow$ AC.

4. Traction Motor & Drive System

- The electric motor provides all traction power for vehicle propulsion and acceleration.
- Power is transmitted to the wheels through a reduction gear.

5. Key Characteristic

- In this configuration, there is no mechanical connection between the engine and wheels.
- The engine only charges the battery, while the motor alone drives the wheels.
- The vehicle can operate in pure electric mode (motor only) or engine-assisted mode (engine charges battery while driving).

Advantages of Series Hybrid Vehicles

1. The series hybrid is a simple system with no complexity in connection.
2. A multispeed transmission system, clutch or converters are not required since the electric motor drives the reduction gear for movement.
3. It works like a pure Electric Vehicle that runs on a battery source.
4. It helps to reduce harmful gas emissions and promotes an eco-friendly environment.

Disadvantages of Series Hybrid Vehicles

1. The series hybrid vehicle propulsion is entirely dependent on the battery source.
2. The energy conversions happening in each stage result in the overall efficiency loss.
3. All these propulsion devices need to be sized to sustain maximum power requirements. In the case of long-grade ability conditions, the vehicle's performance is poor. So a powerful motor is required in this situation, making the Series HEV expensive.

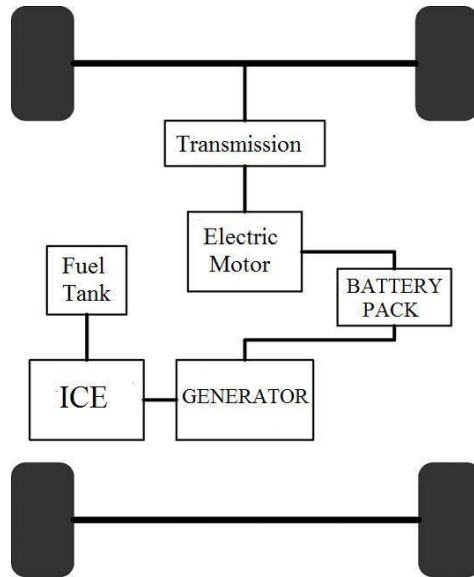


Figure 18. Series type HEV

Parallel types of HEV:

Working Principle

1. Dual Power Sources

- The internal combustion engine (ICE) and the electric motor(s) are both mechanically connected to the wheels.
- Either power source, or both together, can propel the vehicle.

2. Transmission & Power Blending

- A specialized transmission system and clutches are used to combine and manage power flow from the engine and motor.
- The power split between the two sources depends on driving conditions.

3. Control System

- A computerized hybrid control unit constantly monitors speed, load, and driver input.
- It automatically decides whether the engine, motor, or both should provide torque to maximize fuel efficiency and performance.

4. Operating Modes

- Low Speed / Light Load → Motor alone drives the car (quiet, efficient). •
Normal Driving → Engine provides primary power, motor assists when needed.

- *High Load (acceleration, hill climbing) → Engine + Motor work together for maximum torque.*
- *Regenerative Braking → Motor works as a generator, converting kinetic energy into electrical energy stored in the battery.*

5. Key Characteristic

- Unlike a series hybrid, the engine has a direct mechanical link to the wheels.
- This makes parallel hybrids more efficient at higher speeds (less energy lost in conversion).

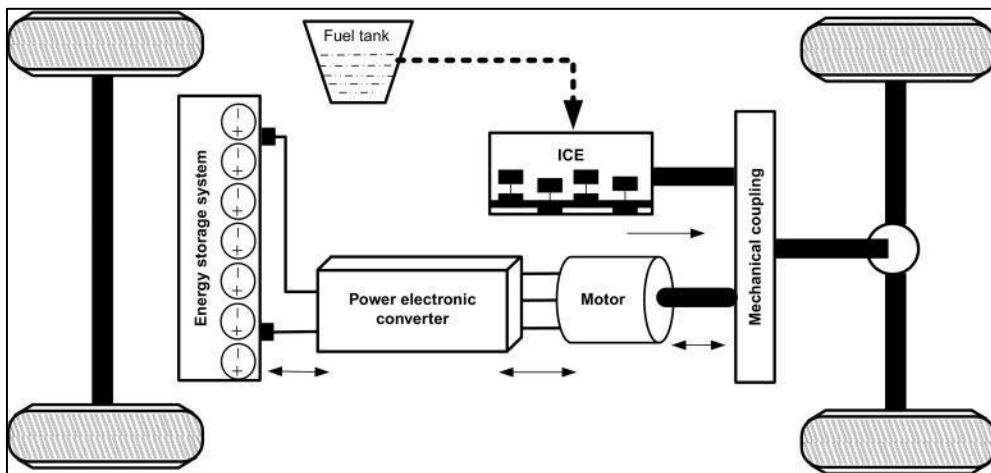


Figure 19. Parallel type HEV

Advantage and disadvantage of hybrid electrical vehicle:

Advantages of Hybrid Electric Vehicles (HEVs):

1. **Fuel Efficiency** – Reduced fuel consumption compared to conventional gasoline vehicles.
2. **Lower Emissions** – Produces fewer greenhouse gases and pollutants.
3. **Regenerative Braking** – Converts braking energy into electrical energy to recharge the battery.
4. **Flexibility** – Can switch between electric motor and internal combustion engine for optimal performance.
5. **Reduced Dependence on Fossil Fuels** – Less fuel is required compared to traditional vehicles.
6. **Smooth Driving Experience** – Electric motor provides instant torque for better acceleration.

Disadvantages of Hybrid Electric Vehicles (HEVs):

1. **Higher Initial Cost** – More expensive than conventional cars due to advanced technology.
2. **Complex Design** – Dual powertrains (engine + electric motor) make the system complex.
3. **Battery Replacement Cost** – Hybrid batteries are costly and may need replacement after years of use.
4. **Limited Electric-Only Range** – Unlike full EVs, most hybrids can't travel long distances solely on electricity.
5. **Maintenance Challenges** – Specialized servicing is required, which may not be available everywhere.
6. **Weight of Vehicle** – Heavier than normal cars due to dual systems and battery packs.

Degree of Hybridization

There are three main categories of hybrid system: micro-hybrid, mild-hybrid, and full-hybrid. A micro-hybrid has an electric motor-generator integrated to provide up to approximately 10% of the maximum engine power in the form of additional torque. A full-hybrid is one in which the electric motor typically provides at least 40% of the maximum engine power as additional torque.

	Engine Start/Stop	Regenerative Braking	Motor Assist	Electric Drive
Micro-hybrid	Yes	Slight	Slight	No
Mild-hybrid	Yes	Yes	Yes	No
Full-hybrid	Yes	Yes	Yes	Yes

1. Micro hybrid

A small motor/generator usually taking the place of a conventional alternator.

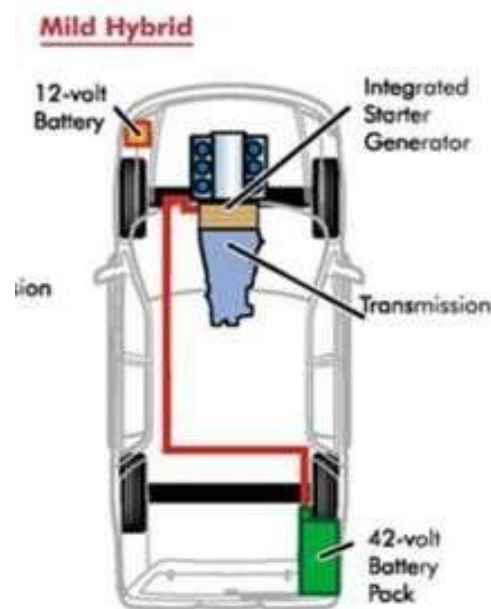
They feature idle-stop function (i.e. automatic turning off the engine when vehicle is not moving).

Utilize Regenerative braking (using kinetic energy of a stopping car to charge the car-battery, by converting its electric motor into a generator or dynamo).

Are not capable of using electric motor alone to propel the vehicle.

Battery is usually 42 volts or less.

Cheaper hybrid system as comparative to medium hybrid.



2. Medium hybrid

A small motor/generator usually taking the place of a conventional alternator. They also feature idle stop feature. Utilize Regenerative braking. Most are not capable of using electric motor alone to propel the vehicle. motor/generator only delivers drive under heavy acceleration or to give the engine a helping hand from a standing start, to ensure fuel economy. Battery is usually 144 to 158 volts. Cheaper than fully hybrid.



3. Full hybrid

They feature idle-stop feature. Fully active regenerative braking. Capable of using the electric motor alone to propel the vehicle. The electric motor also assists the engine. Battery are usually 200-300 volts. Costlier than previous two types of hybrids.

